

Lesson 7: Conditionals Practice

45 minutes

Overview

In this lesson students spend most of their time practicing using the skills and processes they have learned about conditionals. At the conclusion of the lesson students discuss the main things they realized and still have questions about at the conclusion of this lesson.

Standards

Full Course Alignment

CSP Conceptual Framework

- **AAP-2** - The way statements are sequenced and combined in a program determines the computed result. Programs incorporate iteration and selection constructs to represent repetition and make decisions to handle varied input values.
- **AAP-3** - Programmers break down problems into smaller and more manageable pieces. By creating procedures and leveraging parameters, programmers generalize processes that can be reused. Procedures allow programmers to draw upon existing code that has already been tested, allowing programmers to write programs more quickly and with more confidence.
- **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Warm Up (5 minutes)

Quick Warm Up

Activity (35 minutes)

Practice Time

Wrap Up (5 minutes)

Assessment: Check For Understanding: AP Practice

Objectives

Students will be able to:

- Debug programs that use boolean expressions and conditional statements
- Write programs that use boolean expressions and conditional statements with the support of sample code.

Preparation

- Review the programming challenges students will be completing
- Review the Debugging Guide for ideas on how to support your students during the lesson
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Debugging Guide** - Resource
▼ Make a Copy
- **Conditionals Practice** - Slides
▼ Make a Copy
- **Guide to Practice Lessons** - Video

Teaching Guide

Warm Up (5 minutes)

Quick Warm Up

Remarks

Today we're going to have a chance to practice programming with a lot of the concepts and patterns we've explored over the last two lessons. I encourage you to go through these with a partner, but pay close attention to what each other is doing. You will be choosing one of three apps to help develop today. While the design of each of these apps is complete, it will be your job to use your knowledge of conditional statements in order to make the apps function as intended.

Whichever app you choose, please continue to choose that **same** app on each of the choice levels. There are a total of four levels for each app.

In our next lesson you're going to have to use a lot of these skills on an independent project, and these activities are good practice for what you'll find there! Alright, let's get to it!

Teaching Tip

Move Quickly to the Activity: There's a lot in the main activity of today's lesson. You may optionally wish to do a quick vocabulary review or address any questions that came up in the last lesson. Otherwise, give students more time to get hands on with some code.

Activity (35 minutes)

Practice Time

Group: It is recommended that students work in pairs for this lesson. Consider using pair programming, having drivers and navigators switch every 3 minutes, not every level.

 **Do This:** Direct students to Code Studio, Lesson 7 Level 1.



1-7

Conditional Statements with App Development

1

2

3

4

5

6

7



Assessment Opportunity

Create Performance Task Written Response Practice (*Optional*)

In order to help prepare students for the new component of the AP Create Performance Task, there will be several opportunities throughout the curriculum to practice creating written responses aligned with the learning objectives that might be assessed on the exam.

Create PT Learning Objective(s):

- Identify the error.
- Correct the error.

Unit Journal Template You may choose to reuse this debugging journal template if you think it will help support your students becoming stronger debuggers.

Sample Response for Instructors

C
O
D
E

CS Principles

Unit Journal, Unit 4, Lesson 7

Your Voice / Your Choice

Write about a bug you found during your work in class

Describe the problem

- What did you expect it to do?
- What did it actually do?
- Did it always happen?

Write your response here....

Hunt for bugs

- What caused the bug?
- How did you find the bug?

Write your response here....

Try Solutions

- What changes did you make?
- What was the solution you found?

Write your response here....

Wrap Up (5 minutes)

Remarks

Conditionals can be a little bit tricky, but I saw a lot of good progress today in nailing down this concept. We may have a few lingering questions, but you also have a lot of resources available. Next time you'll have a chance to put all this together by programming an app that starts with "the blank screen"!

Assessment Opportunity

Create Performance Task Written Response Practice (Optional)

In order to help prepare students for the new component of the AP Create Performance Task, there will be several opportunities throughout the curriculum to practice creating written responses aligned with the learning objectives that might be assessed on the exam. This activity within the Unit Journal also practices the skill of snipping code segments in order to prepare a Personalized Project Reference for the Create Task.

Sample Prompt: *Find a code segment where you have used logical operators and detail how the outcome of the segment would differ if the logical operations were reversed (i.e., AND changed to OR and vice versa).*

Unit Journal Template

Sample Response for Instructors

Programming Concept: Logical Operators

Snip and paste an example of code using logical operators here!

Code Highlights- Use these colored spotlights to point out & discuss features of this code



Code Highlights



Code Highlight 1

...



Code Highlight 3

...



Code Highlight 2

...



Code Highlight 4

...

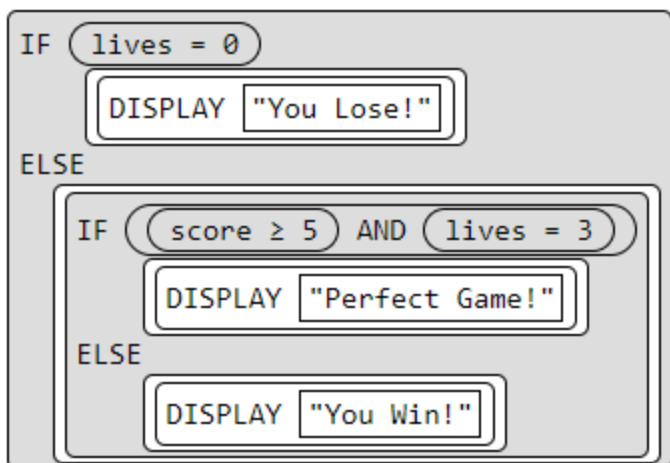
Find a code segment where you have used **logical operators** and detail how the outcome of the segment would differ if the logical operations were reversed (i.e., AND changed to OR and vice versa).

...

Assessment: Check For Understanding: AP Practice

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: What will be displayed after this code segment is run?

`score ← 4``lives ← 3`

Question: The program below asks a user to type in a number and then will output a message. What number will a user need to input for the message "COLD" to be displayed?

```
number <- INPUT()

IF (number >= 10)
{
  IF (number <= 20)
  {
    DISPLAY("MEDIUM")
  }
  ELSE
  {
    DISPLAY("HOT")
  }
}
ELSE
{
  DISPLAY("COLD")
}
```

☰ 8-9

Check For Understanding: AP Practice



This work is available under a **Creative Commons License (CC BY-NC-SA 4.0)**.

If you are interested in licensing CodeAI materials for commercial purposes **contact us**.